

ASSENS TOROE, Denmark

WMO: 061230

Lat: 55.2444N

Lon: 9.8883E

Elev: 10

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	22.5	24.9	16.5	12.6	24.6	19.2	14.4	26.9	35.8	40.4	32.6	41.0	10.1	70	0.689

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.4	77.8	67.0	74.8	65.6	71.9	64.1	68.6	75.2	66.8	72.6	65.2	70.5	11.1	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.9	95.7	72.4	64.3	90.4	70.4	62.8	85.6	68.3	32.8	75.4	31.3	72.8	30.1	70.6	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 31.2	27.5	24.7	DB	20.1	83.1	4.4	2.9	17.0	85.2	14.5	86.9	12.0	88.6	8.9	90.7	(4)
(5)			WB	19.0	70.9	4.2	2.0	16.0	72.3	13.6	73.5	11.2	74.6	8.2	76.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.5	35.5	35.3	39.0	46.6	54.3	59.7	64.4	64.4	59.3	51.5	44.4	39.1	(6)	
	DBStd	11.59	5.93	5.14	5.22	5.06	5.32	4.29	4.71	3.80	3.91	4.75	4.91	5.74	(7)	
	HDD50	1898	450	412	341	127	16	0	0	0	0	40	174	338	(8)	
	HDD65	5748	915	832	806	551	335	167	72	56	176	418	617	803	(9)	
	CDD50	1733	0	0	0	26	149	292	445	447	279	87	8	0	(10)	
	CDD65	105	0	0	0	0	2	8	52	38	5	0	0	0	(11)	
	CDH74	326	0	0	0	0	7	21	201	92	5	0	0	0	(12)	
	CDH80	28	0	0	0	0	0	0	21	7	0	0	0	0	(13)	
(14) Wind	WSAvg	12.7	14.5	13.3	12.1	10.9	11.1	12.1	11.6	11.8	12.7	13.5	14.5	14.8	(14)	
(15) Precipitation	PrecAvg	29.1	2.8	2.0	1.8	1.4	2.0	2.5	2.7	3.1	2.8	2.9	2.6	2.7	(15)	
	PrecMax	37.0	5.8	4.8	4.2	3.5	3.3	5.6	5.9	6.2	5.8	6.9	5.4	5.7	(16)	
	PrecMin	20.0	0.1	0.3	0.2	0.1	0.6	0.0	0.5	1.1	0.5	1.1	0.6	0.7	(17)	
	PrecStd	4.6	1.4	1.0	1.0	0.7	0.9	1.2	1.4	1.5	1.4	1.1	1.2	1.3	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.5	46.9	55.1	64.9	74.4	77.2	82.7	80.7	74.0	63.2	56.6	50.3	(19)	
		MCWB	46.4	44.3	48.0	53.9	63.1	64.9	68.3	68.1	64.5	59.1	54.5	48.5	(20)	
	2%	DB	46.5	44.8	49.6	60.9	69.7	72.9	79.0	76.5	69.9	60.6	54.2	48.6	(21)	
		MCWB	44.7	42.7	44.8	51.8	59.1	63.2	67.3	66.9	62.9	57.3	52.4	46.9	(22)	
	5%	DB	45.0	43.3	47.0	57.5	66.1	70.0	76.0	73.6	67.3	59.1	52.4	47.4	(23)	
		MCWB	43.3	41.4	43.4	50.8	58.3	62.3	66.4	65.4	61.4	56.3	50.6	45.8	(24)	
	10%	DB	43.4	41.9	45.0	54.3	63.0	67.2	72.7	70.7	65.2	57.7	50.9	46.1	(25)	
		MCWB	41.8	40.2	42.3	48.8	56.5	60.7	65.2	64.0	60.3	55.2	49.0	44.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.0	44.9	48.4	55.6	65.1	67.5	71.5	70.2	66.2	60.1	55.0	49.1	(27)	
		MCDB	47.8	46.0	53.7	62.3	72.8	73.0	79.0	77.3	71.5	62.2	56.3	49.8	(28)	
	2%	WB	45.0	43.3	45.8	53.2	61.5	65.4	69.3	68.1	63.8	58.2	52.7	47.3	(29)	
		MCDB	46.1	44.5	49.1	58.9	67.3	70.8	76.2	74.8	68.1	60.0	53.9	48.3	(30)	
	5%	WB	43.5	41.7	44.0	51.2	59.2	63.3	67.4	66.4	62.5	56.8	50.9	46.1	(31)	
		MCDB	44.7	43.0	46.1	56.1	64.4	68.1	73.7	71.7	66.2	58.5	52.2	47.2	(32)	
	10%	WB	41.9	40.4	42.7	49.4	57.0	61.5	65.6	64.8	61.1	55.6	49.3	44.9	(33)	
		MCDB	43.3	41.7	44.6	53.7	62.3	65.9	71.5	69.6	64.6	57.4	50.7	46.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.1	5.4	7.7	10.6	11.2	10.9	10.7	9.4	8.3	6.3	5.0	5.0	(35)	
		MCDBR	5.3	6.6	10.7	15.6	16.3	15.7	16.0	14.2	11.3	7.2	4.7	4.8	(36)	
	5% WB	MCWBR	5.3	5.4	7.1	9.2	9.8	9.5	8.2	7.4	6.8	5.4	4.7	4.7	(37)	
		MCWBR	5.4	6.3	9.4	14.0	15.0	13.7	14.4	12.4	10.0	6.5	4.7	4.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.320	0.340	0.366	0.399	0.392	0.387	0.404	0.404	0.381	0.358	0.337	0.307	(40)	
		taud	2.268	2.295	2.270	2.229	2.320	2.363	2.331	2.344	2.390	2.399	2.329	2.255	(41)	
	Ebn at Noon		188	230	252	260	270	272	264	257	246	222	181	163	(42)	
		Edh at Noon	18	25	32	39	37	36	37	34	29	23	17	15	(43)	
(44) All-Sky Solar Radiation	RadAvg		154	349	777	1299	1675	1774	1655	1328	934	482	193	112	(44)	
	RadStd		25	51	72	152	165	134	180	94	95	59	28	10	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (56 neighbors)		+0.63	N/A	N/A	N/A	N/A	N/A	N/A	-277	N/A	N/A	

Nomenclature: See separate page